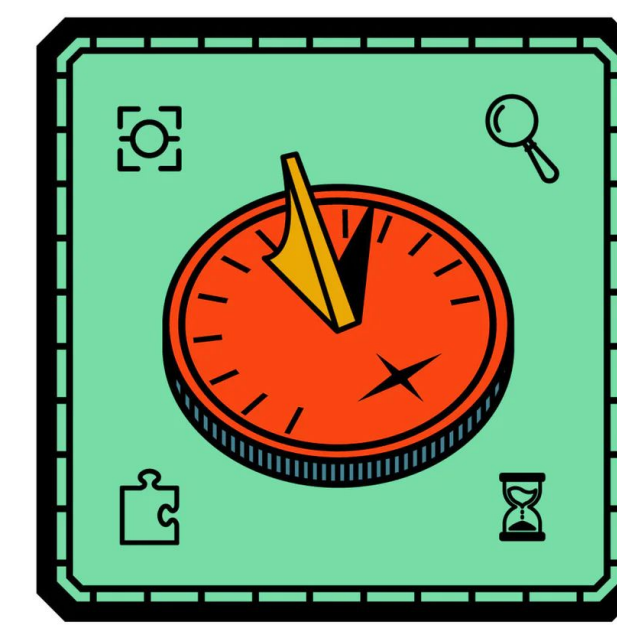




DECODE



Mamaroneck Robotics 8490

Engineering Portfolio

2025-2026

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Meet The Team



Team Members:

Mustafa Khan, Finn Schweter, Luke Minchilli, Hugh Scobie, Micah Robbins, Jordan Metsch, Ethan Bregman, Violet Barber-pertuch, Brian Orozco Miranda, Max Prager, James Abend, Misha Vaculin, Harper Morgan, and Ari Steinberg

About Us:

Team 8490 represents Mamaroneck High School, in Mamaroneck, New York. The team started competing in the FIRST Tech Challenge in 2015 and since then, our school has implemented an impressive engineering program, and our team has grown with it.

The team meets several times a week where we design, program, and troubleshoot.

To better organize ourselves and improve efficiency, we split the team into three branches, though all members are welcome to move and help with whatever is needed. Our Engineering branch is responsible for designing, modeling, and building the robot, while Coding focuses on the software aspects. Documentation and Outreach handles community engagement and records.

While we have had our challenges through the years, our team has always persevered, and we look forward to challenging ourselves and growing our skills in the future.

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Team Goals

2025-2026 Goals

For this FTC year, we wanted to focus on designing a ever evolving robot, and providing learning opportunities for members new and old. We wanted to be unique in our robot design and we can credit that to our constant divergent thinking and repeated converging to evolve and improve our attachments. The consistent

We have a good mix of seasoned veterans and new participants this year creating a perfect opportunity for them to pass on the torch and keep improving our teams skills and providing new learning opportunities for all. In addition, our team hopes that our continued participation in the FTC competitions will grow our skills and knowledge as engineers and programmers. We have been looking to make a positive impact on our community, grow more united as a team, and have fun throughout the process.

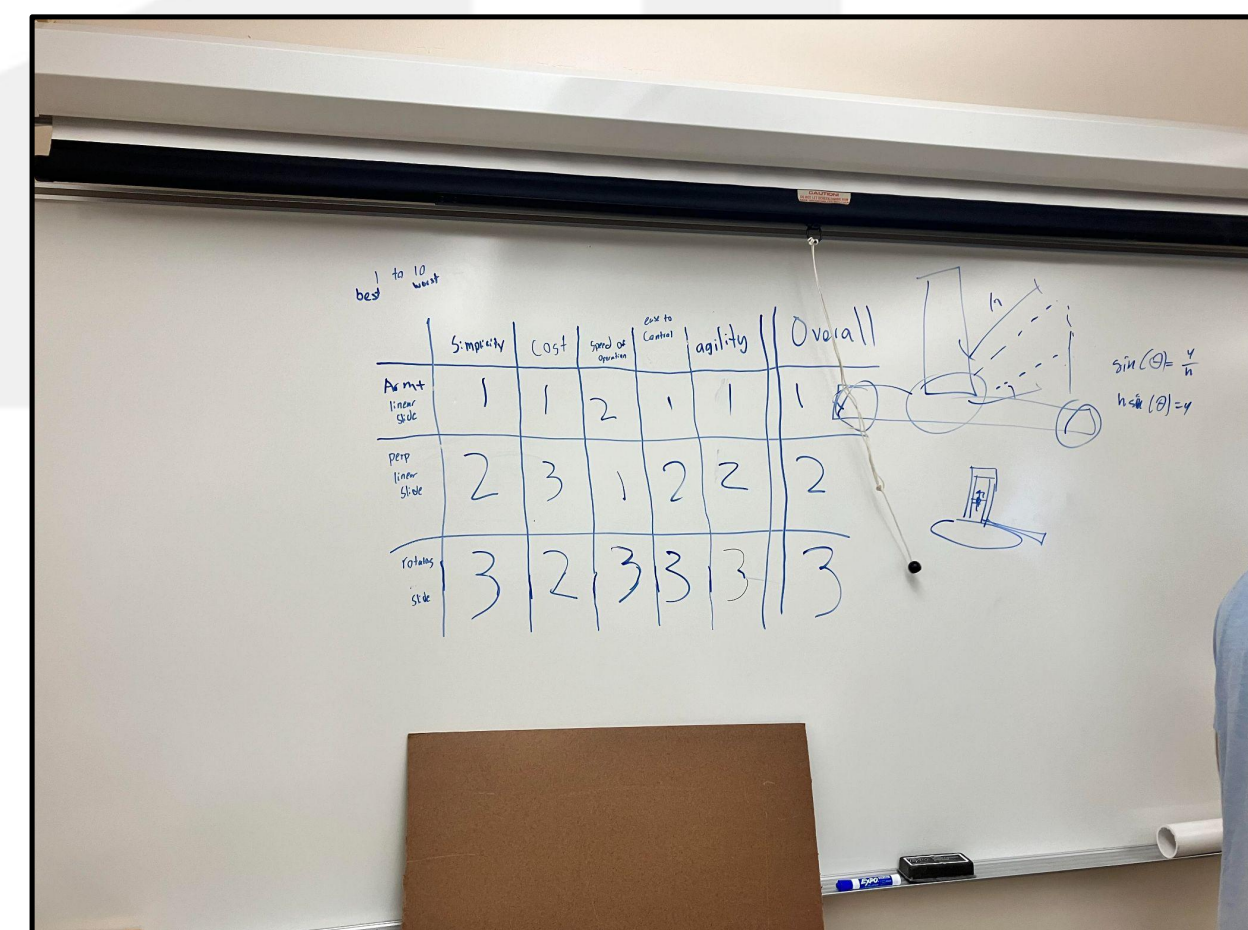
Design Process

Our design process began with **divergent thinking** - we split into small groups to brainstorm, and then came back together to discuss. We discussed theory about how our robotics decisions would affect the robot's actions. By following this process, our team was able to complete our goals in an efficient and effective manner. During meetings, we discussed possible intake and launcher designs.

We used a decision matrix as a group to determine what to use for an intake and launcher. This matrix included different categories which would be ranked on a scale from one to five.

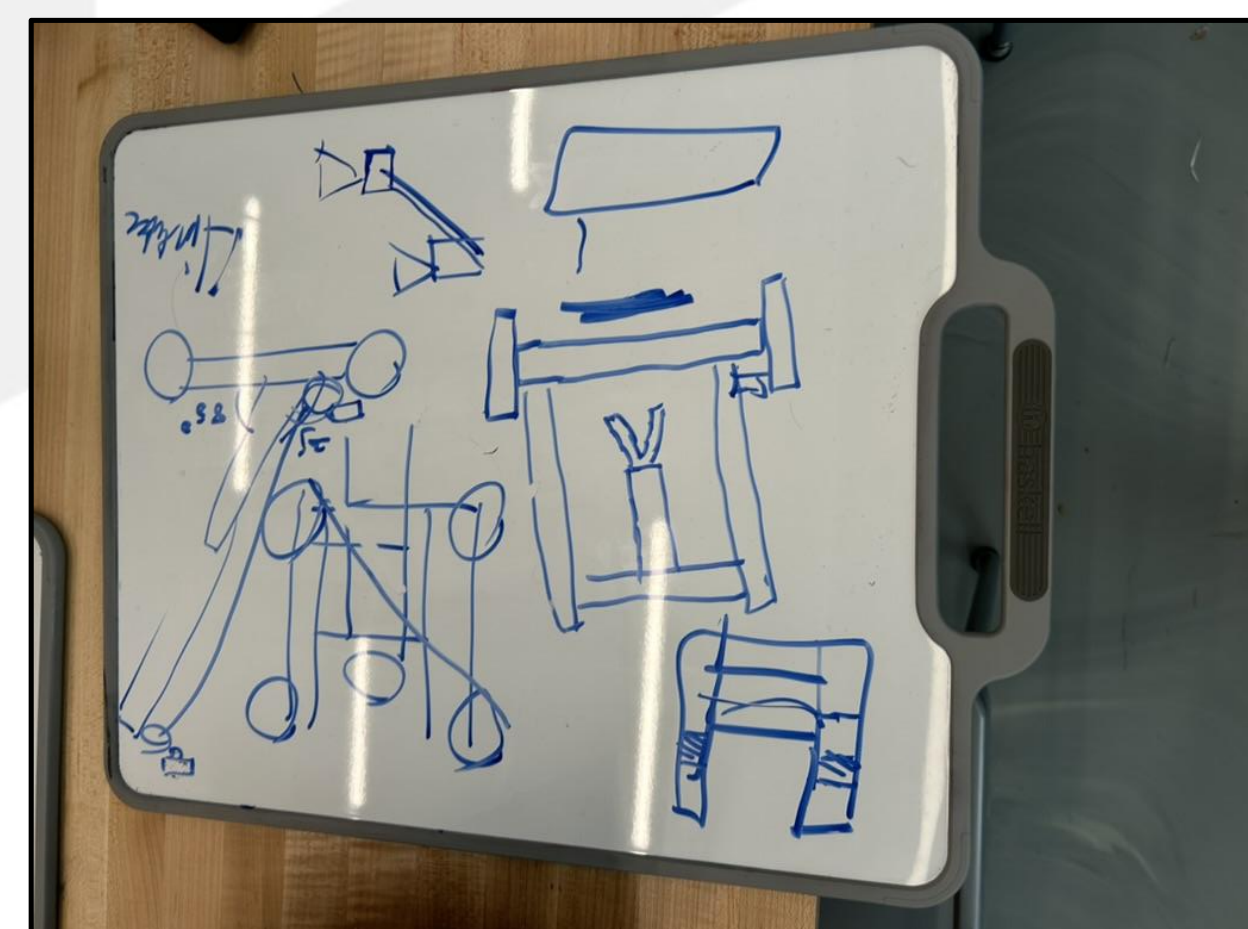
Another challenge was how we could attach the servo on the end of the wrist motor. Because this motor is set back from the actual pivot point, there was no good way to put the servo on the end. Eventually, we realized we could use a belt with a tensioner, which would allow us to fix it if it stretches out.

Despite the challenges, each new iteration is valuable in constructing our robot, and we make sure to enjoy the entire process.



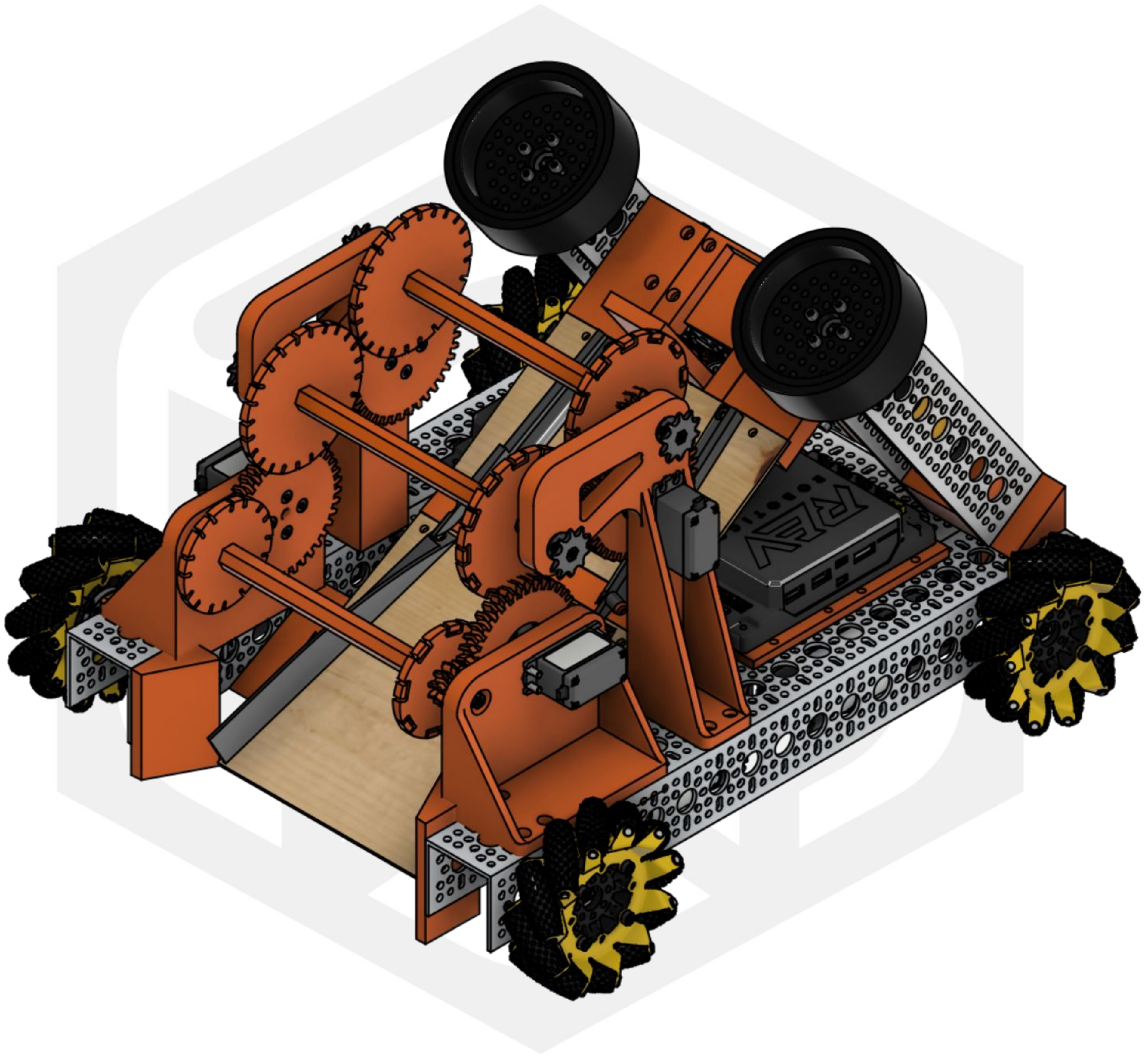
	Simplicity	Cost	Speed of Intake	Overall
Design 1	1	1	2	1
Design 2	2	3	1	2
Design 3	3	2	3	3

The team's rankings of different intake designs



One of our early ideas for the robot

Final Design



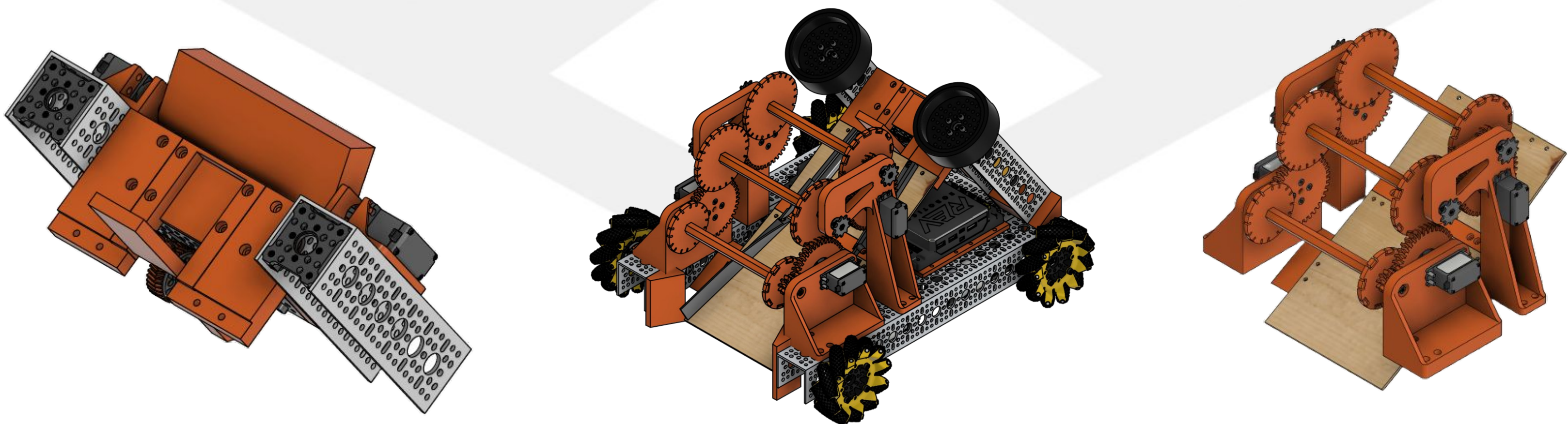
Custom Parts

Why custom parts?

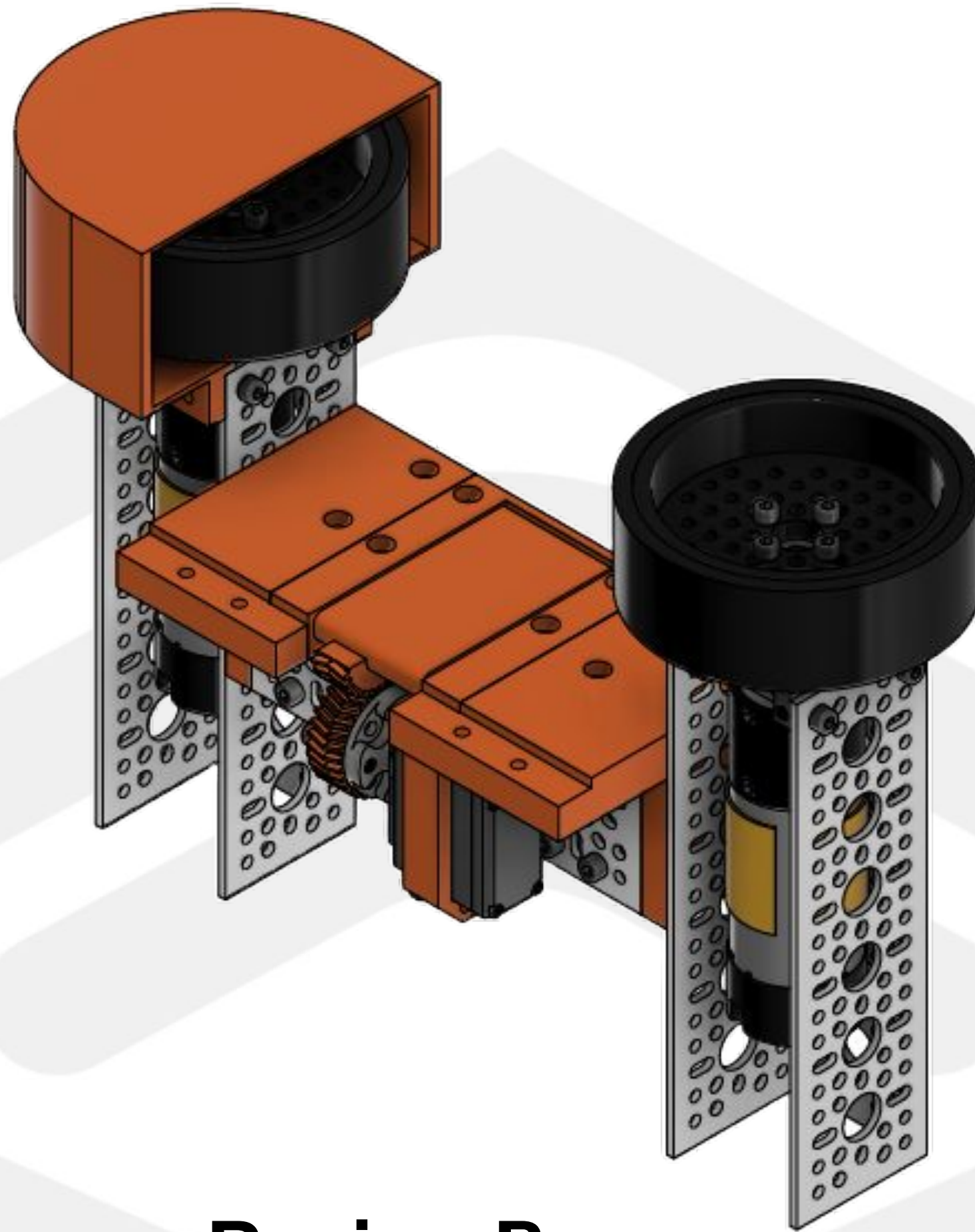
As a school club, we need to make do with what we have available to us, as we have limited funds and time.

Off-the-shelf parts do not meet the specific needs our robot has to succeed in the season's challenges. We heavily use 3D-printed parts designed with Fusion 360 to both optimize our robots layout and compete in competition tasks. Our most notable 3D printed parts are a combined battery and control hub mount, spacers for linear slides, a belt tensioner, and our claw. Another 3D printed part are our custom side panels which attach to the robot using magnets. We also needed to build our own cable for the arm because we could not find a premade one of the right length that was coiled the way we need. To do this we ran smaller wires through heat shrink and coiled it by heating it around a pipe. These parts both help us be more adaptable throughout our robot's design process and make us stand out.

CAD Models



Launcher



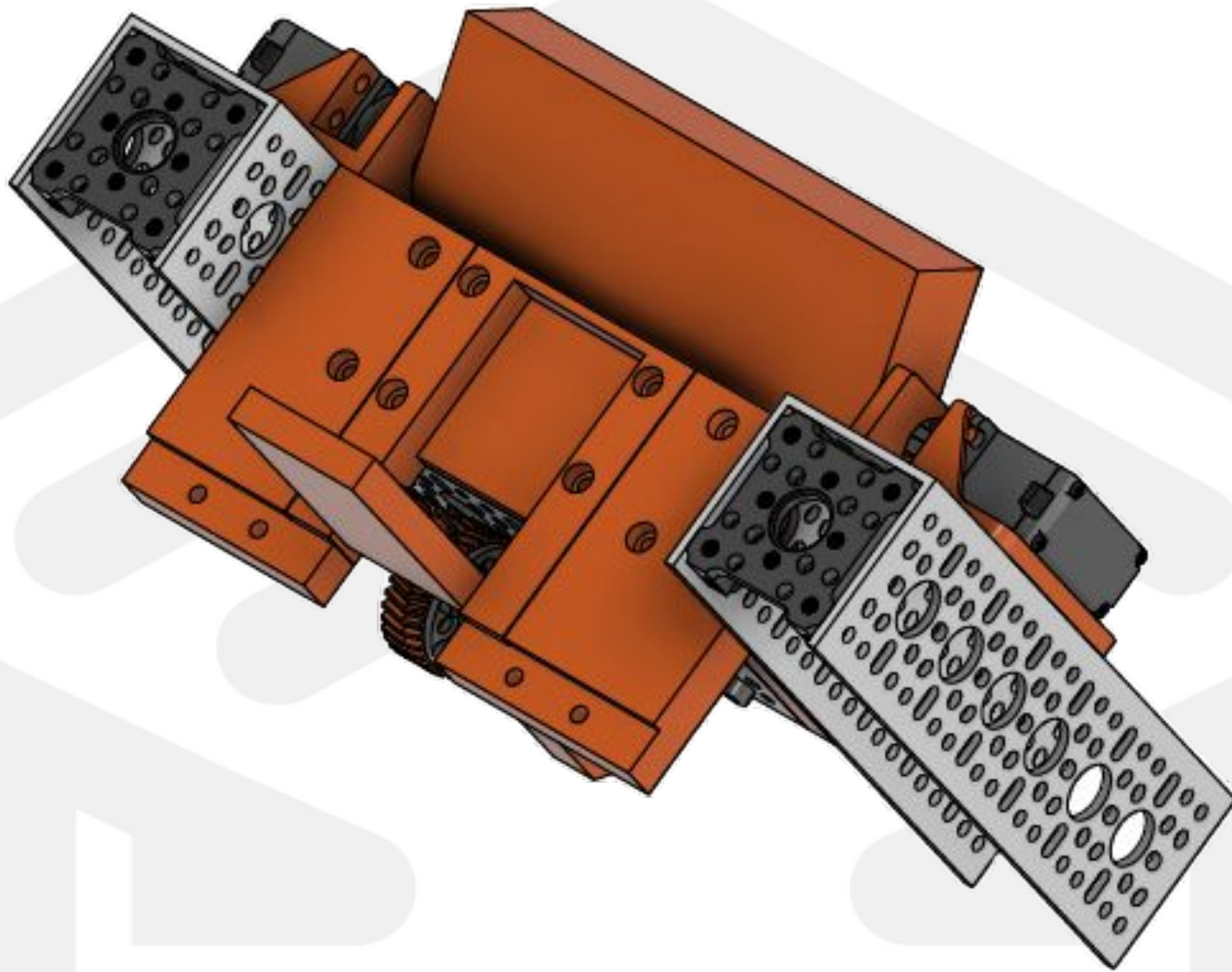
Design Process

As a school club, we need to make do with what we have available to us, as we have limited funds and time. Off-the-shelf parts do not meet the specific needs our robot has to succeed in the season's challenges. Early on we had a basic idea about how we wanted to design a subsystem to score artifacts. Right from the beginning we had planned to use flywheels to score the artifacts. As a team we discussed whether we should use a hooded style flywheel with one wheel or a dual flywheel. We created a matrix and weighed the pros and cons of each design and eventually decided on a dual flywheel system because we deemed it to be more reliable and consistent. After running through many different iterations of our launcher, we eventually created a working design. Once we had our design, we tested it in conditions similar to a real match in order to ensure that it has good reliability, accuracy, and consistency. Through this process of design, problem-solving, and resourcefulness, we were able to create an intake and transfer mechanism that was able to work effectively.

Specifications

- 13.7:1 ratio gear 5203 Series GoBilda Yellowjacket motor
 - Intakes and outtakes on opposite sides to save time
 - Rotates 150°

Gate and Ramp



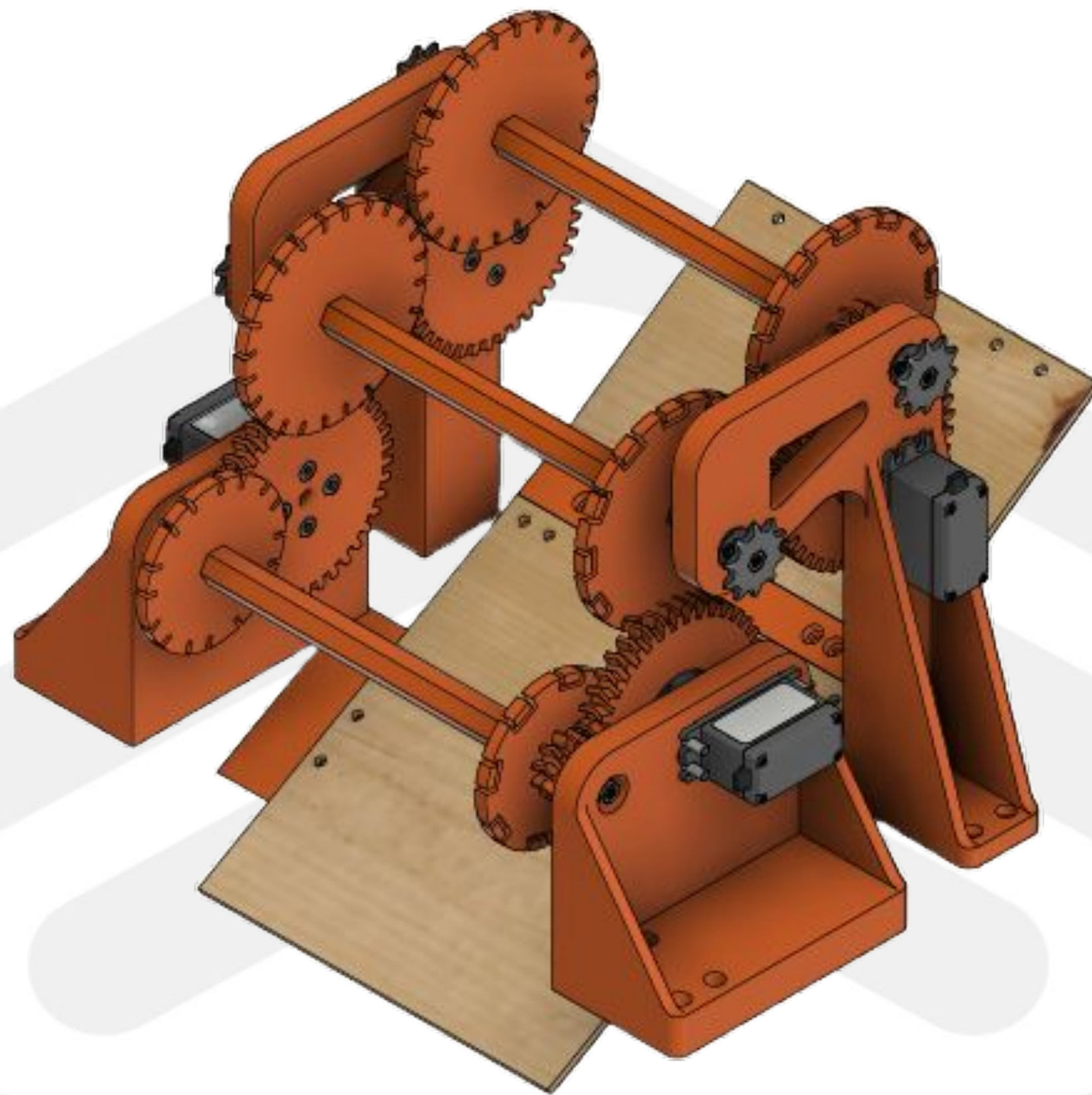
Design Process

We wanted to be able to hold more than one artifact so that way we were able to pick up more artifacts and score more efficiently. However we noticed that the artifacts would be picked up the intake and right into the shooter when it wasn't ready to fire. Leading us to create that when used would activate a servo to lift the gate and that would prevent the artifacts from going into the shooter till we were ready to fire. The ramp was created when we wanted to be able to shoot the artifacts at multiple distances around the field. We then 3D printed components that would attach to the launcher at both ends and hold the torque servos that were connected by a 3D printed ramp that when activated it would raise the bar up and done creating different angles for us to shoot the artifacts.

Specifications

- 3 Gobilda torque servos
- 3D printed Herringbone gears

Intake / Transfer



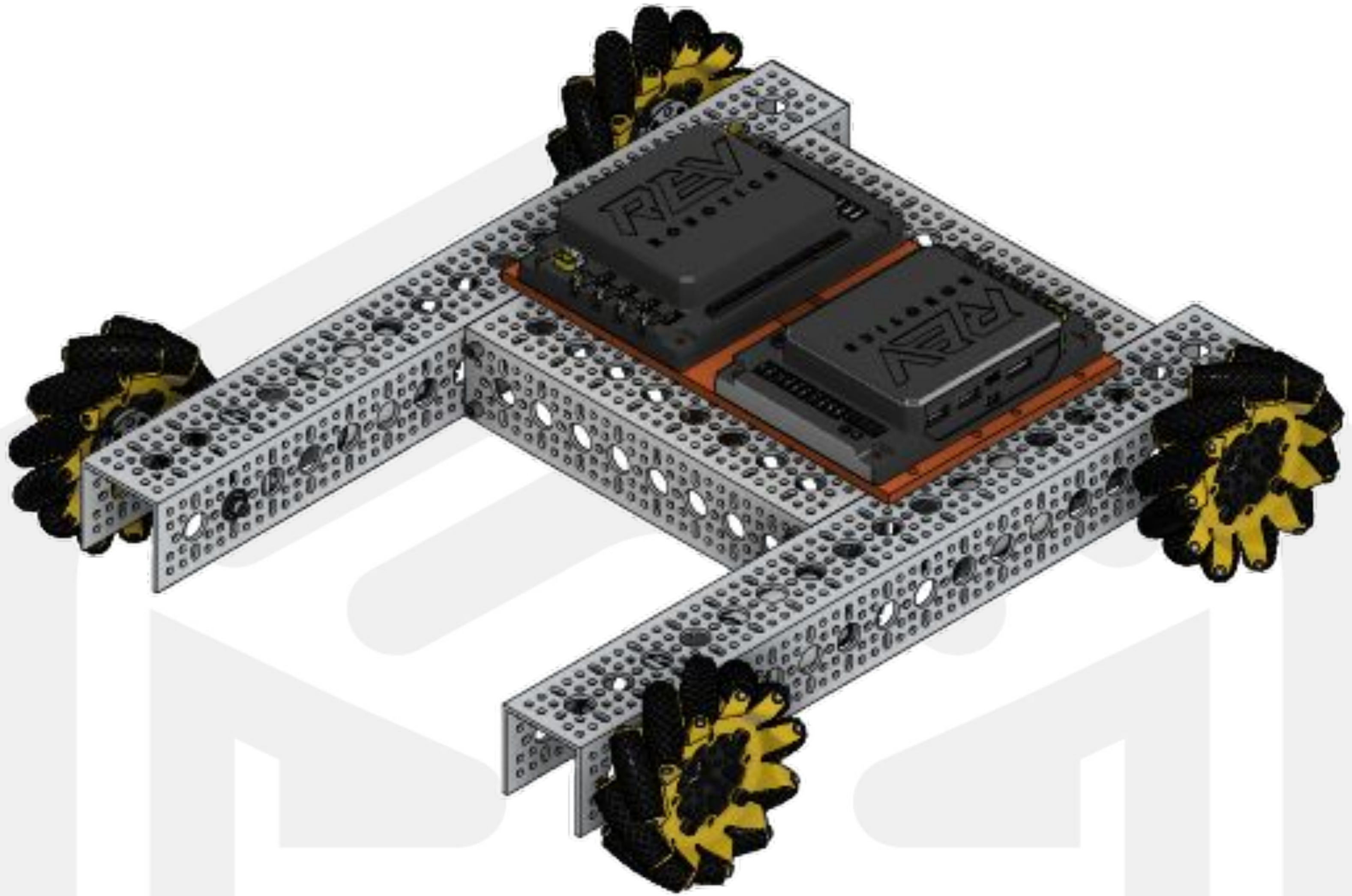
Design Process

The design process for our robotics team's intake and transfer involved several stages of brainstorming, prototyping, and iterative refinement. Our initial prototype consisted of a surgical tubing style intake, but we quickly realized that this style of intake was not very efficient or effective. We discussed with other teams as well as tried alternative styles to intake the artifacts. We tested a similar intake, however instead of using surgical tubing, we decided to try zip ties because they have a greater stiffness and are able to have a greater density. However, this didn't work as expected. We eventually decided to move onto a rubber band style intake and transfer. We tested different amounts or rubber bands, size of wheels, speeds, tension of rubber bands, etc. We eventually figured out the optimal specifications to make the intake and transfer have the greatest effectiveness and efficiency. We also employed 3D printing which allowed us to create custom components to our exact specifications. We decided to use a set of geared servos to control our intake and transfer because we determined that it gave us greater accuracy while maintaining a smaller footprint which makes packaging easier. It also had more than a sufficient amount of speed. Once we decided on our final intake and transfer design, we tested it under match like conditions in order to determine its reliability, accuracy, and consistency. The final intake and transfer design demonstrated that it was able to intake artifacts from multiple orientations as well as a reliable transfer into the flywheels. Through this process of design, problem-solving, and resourcefulness, we were able to create an intake and transfer mechanism that was able to work effectively.

Specifications

- 4x Gobilda Speed Servo
- 3D printed herringbone gears

Chassis



Design Process

As with previous years, we built the chassis primarily with GoBilda U-Channel. This year, we wanted to maximise space on top of the chassis so we decided to mount the motors in the U-Channel. Because we had extra space under our ramp, we decided to mount our electronics underneath to conserve space for the numberplates and intake.

Specifications

- Chassis dimensions 45.5cm x 40.6cm
 - 96mm GoBilda mecanum wheels
- 5203 planetary gear series GoBilda yellow jacket motors with a 19.2:1 gear ratio and 312 rpm

Outreach

Outreach allows our team to give back to our community, make connections with local organizations, improve community knowledge of technology and inspire others to pursue interests in STEAM.

Mamaroneck Repair Cafe



Our team contributed significantly to the Mamaroneck Repair Cafe by assisting with repairing primarily electronic devices. It was an extremely rewarding experience to give back to the community and reduce e-waste, all using the skills we spend hundreds of hours developing during robotics. We hope to continue to assist and potentially host our own repair cafe in the future.

Green Budget Challenge

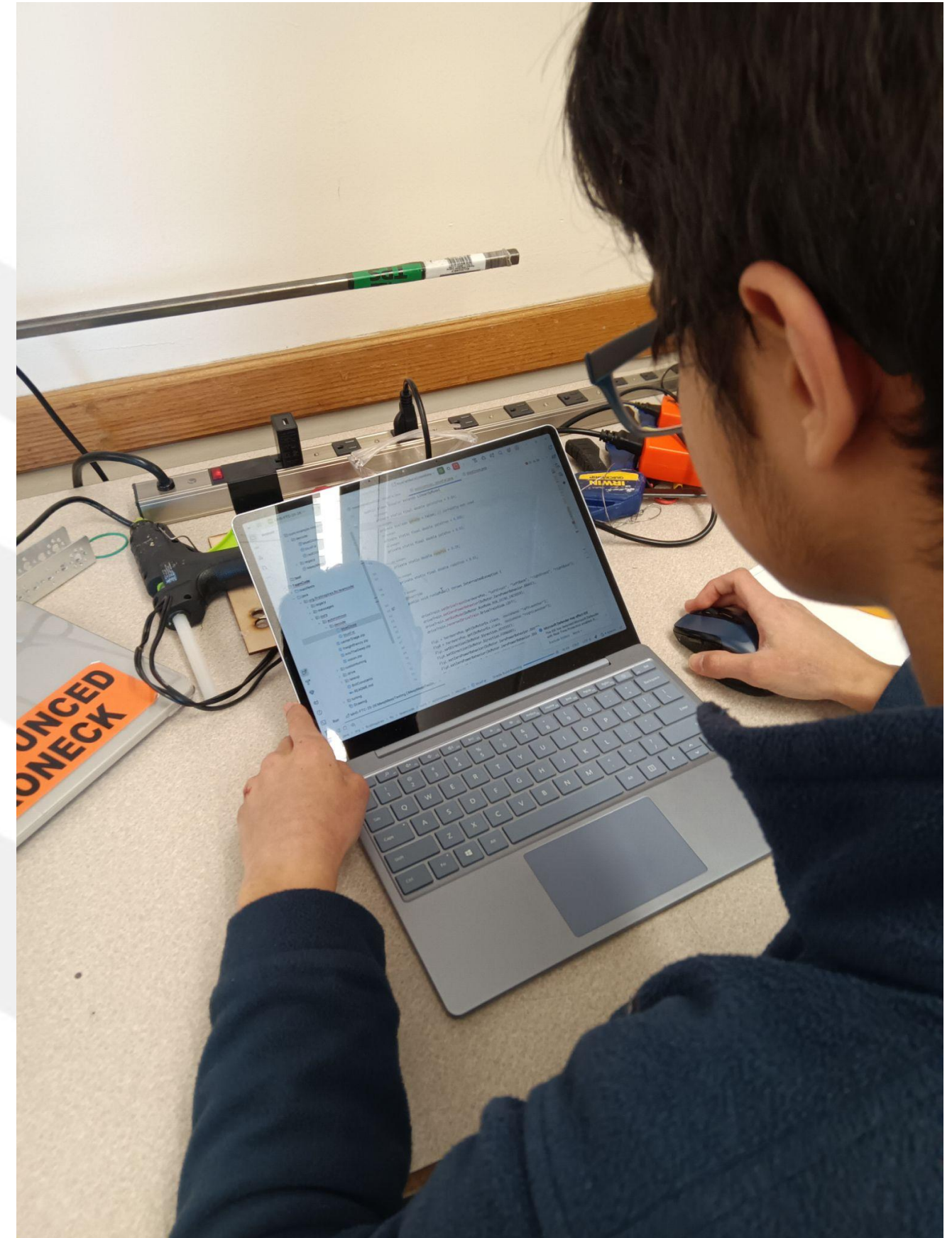


The Green Budget Challenge was an event where we collaborated with the Economics of the Sustainable Energy Club to teach middle school students the trade offs in renewable energy. During the FTC offseason our team participates in MATE, an underwater robotics competition which focuses on sustainability in technology. Our goals of sustainability were matched perfectly with the ideas demonstrated during Green Budget Challenge.

Programming

Goals

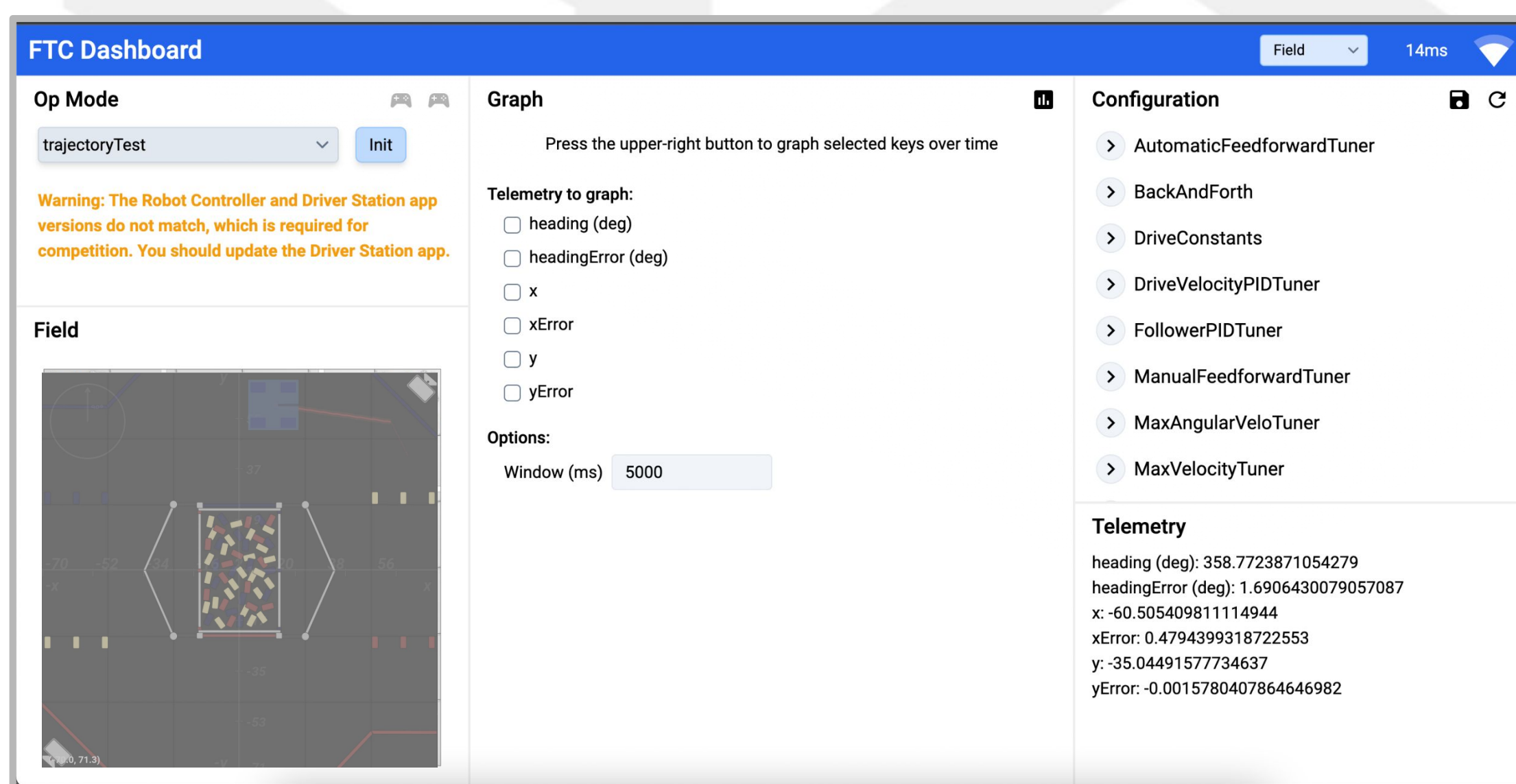
When the programming team began coding the robot, they had two goals in mind: ease of operation and versatility. The most important thing for the success of our team was that the robot had to be easy to operate for any driver. From the start, we had ambitious goals of meeting every task in the challenge, and the coding team made sure that every need was met. To strengthen the ease of operation, the programming team included many driver enhancements such as a toggle button for the intake, intuitive controls, and automatic positioning for the outtake.



Misha works tirelessly on the computer vision programming

How It Was Done

For both the TeleOp and autonomous portions, the robot was coded in Java using Android Studio. The autonomous portion utilized roadrunner in order to make easy and accurate paths for the robot to



Roadrunner was used for the autonomous phase

follow during the autonomous phase of the competition. We also used MeepMeep to draft our autonomous paths before transferring them to code that the robot uses.

TeleOp

```
while (opModeIsActive()){

    telemetry.addData( caption: "X", -gamepad1.left_stick_y);
    telemetry.addData( caption: "Y", gamepad1.left_stick_x);
    telemetry.update();

    move(gamepad1.left_stick_x, -gamepad1.left_stick_y, gamepad1.right_stick_x, speed);

    if(gamepad1.right_bumper){ // intake motor control
        intake.setPower(1);
    }else{
        intake.setPower(0);
    }

    if(gamepad1.left_bumper){ // spinner motor control
        spinner.setPower(.65);
    }else{
        spinner.setPower(0);
    }

    if(gamepad2.dpad_down){
        OuttakeLowerPos(telemetry, gamepad1);
    }

    if(gamepad2.dpad_right){
        OuttakeMiddlePos(telemetry, gamepad1);
    }

    if(gamepad2.dpad_up){
        OuttakeUpperPos(telemetry, gamepad1);
    }

}
```

The main TeleOp Code

Code

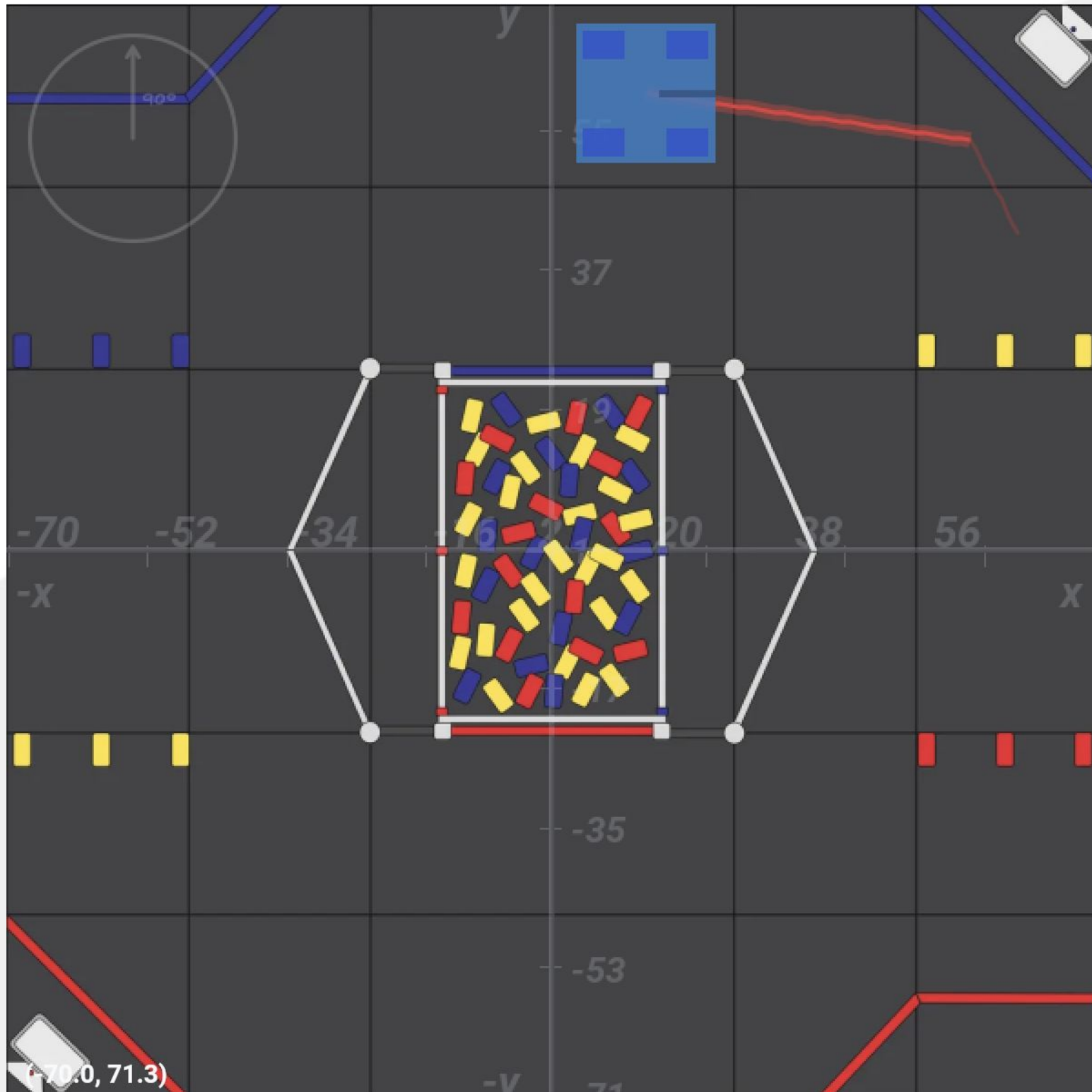
The TeleOp code was designed with scalability and readability in mind. The code has several custom classes such as the drivetrain class which allows all drive motors to be controlled by one line. The code also has several robust methods that control our flywheels, intakes and ramps. In order to effectively control our intake mechanisms, we programmed 2 servos synonymously to deliver effective torque and precision.

Driving

The team focused on minimizing driver workload while maintaining precise control. Controller inputs were designed to be intuitive and consistent, allowing the driver to focus on the field rather than worrying about their controller inputs. We added multiple quality of driving improvements such as slow mode and ramp adjustments. Frequently repeated actions were automated through single-button controls.

Autonomous

Movement



Meep Meep is a program that visualizes Road Runner paths

The team decided on using the Road Runner library to program autonomous movements because of its convenience and ability to program complex paths. Using data from PID loops, motor encoders and the IMU, the robot is able to follow the paths very precisely. This year, the autonomous code aims to shoot the initial 3 Artifacts into the goal and intake 2 more sets of balls to be shot.

Actions

Differing from any other year, we decided to use RoadRunner 1.0's newly added actions to improve our autonomous sequence. We made actions for nearly every aspect of our robot: flywheel spin, intakes and ramp movement. Separating these robot functions into actions makes the code easier to read and write which is especially useful for a robot that has many different moving parts.

Challenges

Challenges This Year

New Membership

One of our most significant challenges this year was the sudden drop of experienced seniors. During the 2024-2025 season our team was made up of primarily seniors that had been in robotics since freshman year.

With the beginning of this year's season we put in our best efforts to increase outreach and gain new members to carry the torch. This issue, although significant, actually allowed us to improve the knowledge of all members due to the trial and error, and pure workmanship of our team



Teaching an FLL member about our robot

Lesson 0: Install softwares

Lesson 1: Overview of a robot

- Chassis
 - Types
 - Advantages
 - Us
- Intake
- Outtake
- Grabbing
- Controlling

Lesson 2: CAD Basics 1 - basic sketching, extruding

- How fusion works
 - 2D sketch → 3D object
 - Planes
 - Toolbar
 - Bodies
 - components

The curriculum plans for new members

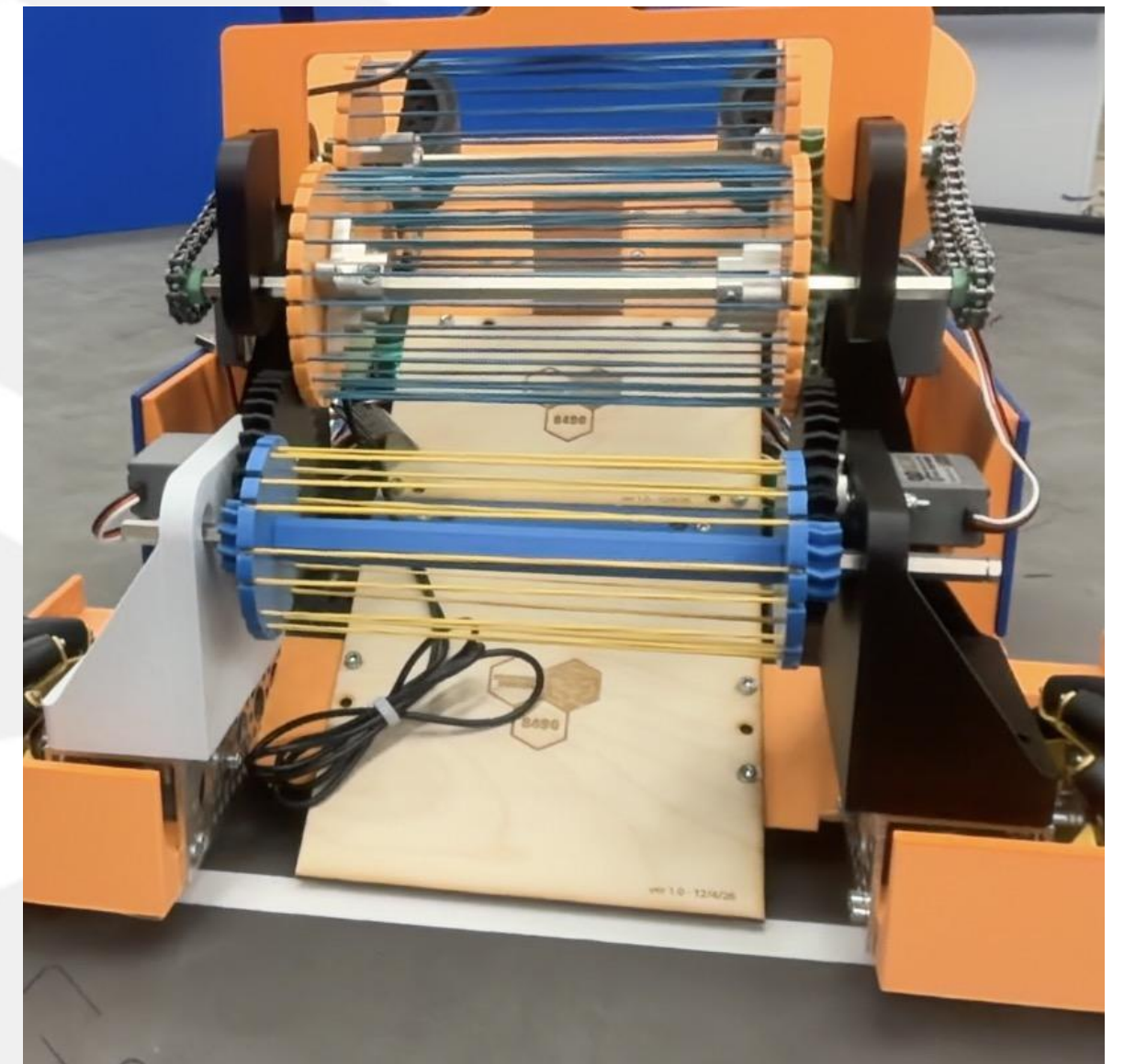
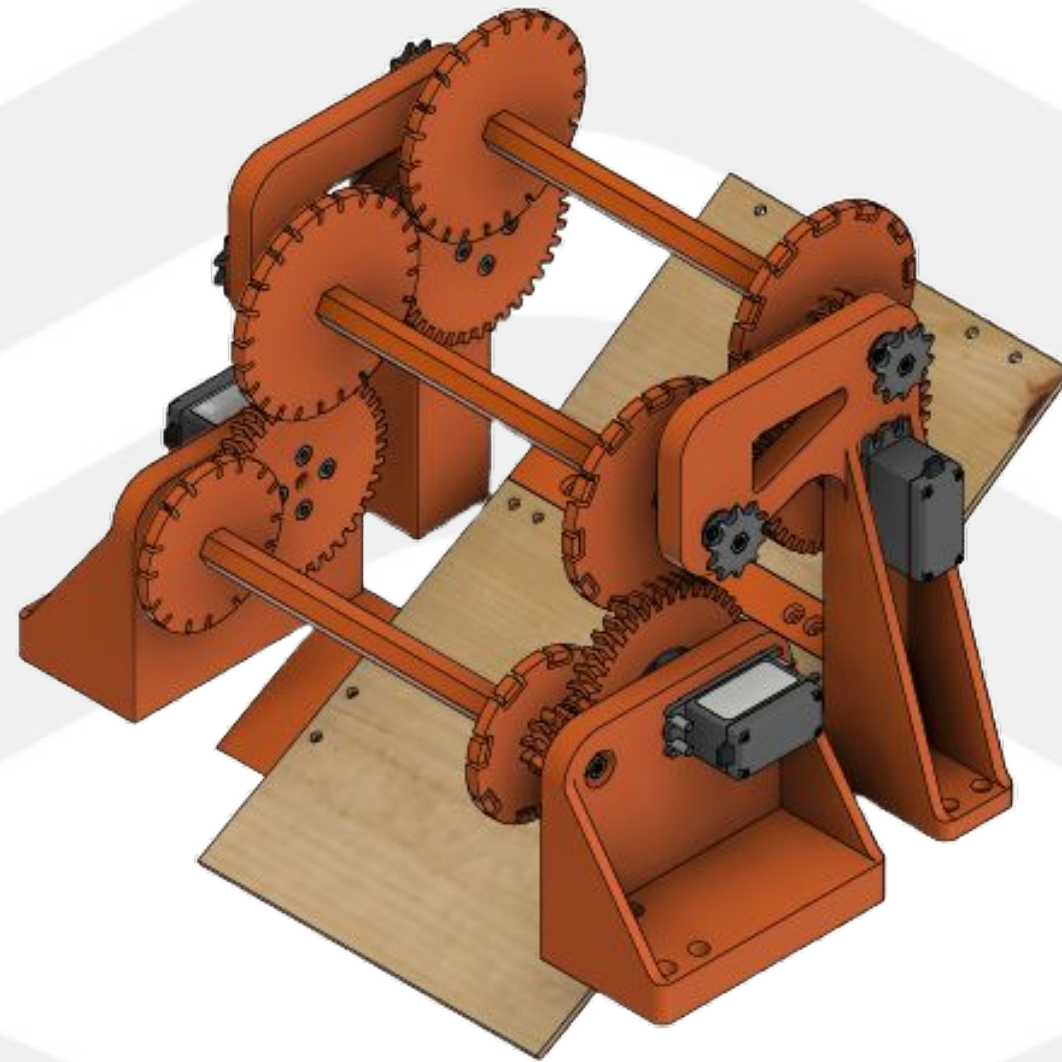
Organization

Throughout this year's season, an issue we came across was lack of organization regarding located materials for engineering, and managing software. For example, our robotics classroom was very disorganized, resulting in time wasted looking for simple tools, that could have been used putting our thoughts to work. To satisfy this task, we made it goal of organizing materials, like our nuts, bolts, and screws. Also, after continuing to manage software, we decide to keep track of tasks accomplished.

Iterations

Prototypes

These were the progressing versions of our intake system.



From the beginning of our design process, the intake was something none of us were really quite sure on how to begin. We had several ideas, but deciding on one specific design was difficult. Originally, we were going to use plastic rods which were rotating on a shaft to grab ahold of the objects in the game arena, but we decided to take a different approach and use rubber bands on our intake system. We decided that this was generally more efficient because the rubber bands had a better grip of the objects from afar. We decided to implement the rubber bands via rotating rollers, but we ended up having to make size changes and add more in quantity for better reach and locomotion.